Dynamic Equilibrium Events in Coupled Toroidal-Helical Transport: A Weak-Field Framework for Finite-Separation Equilibria

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This manuscript distinguishes derived results, numerical experiments, and hypotheses. The full nonlinear Einstein-matter evolution remains future work.

Abstract

We formulate a coupled radial-response and toroidal-helical transport model intended to test whether a finite-separation equilibrium can arise dynamically rather than being imposed as a kinematic constraint. The model is built from a covariant action containing a nonlinear radial scalar, two transport phase fields, and a field-dependent kinetic mixing. We define a Dynamic Equilibrium Event (DEE) as a finite, locally stable stationary point of the radial sector under transport backreaction. For a minimal polynomial realization, we obtain a closed-form finite-separation branch and show that its radial stability follows directly from the equilibrium condition. We derive the matter stress-energy tensor and prove that its null contraction is nonnegative for every null vector whenever the two-field kinetic matrix is positive definite. We then study an explicit non-oscillatory transport configuration and solve the cylindrically symmetric linearized Einstein equations for a localized source. After a positive source rescaling of 10^-6, the largest metric perturbation is of order 10^-3, while the tested co-rotating and counter-rotating null-energy contractions remain positive. Exploratory reduced-metric calculations are also reported to show why prescribing off-diagonal geometry without a compatible matter source can generate energy-condition violations. The present result is a consistent weak-field construction and a set of nonlinear initial-data targets; it is not yet a proof of global non-collapse in the full Einstein-matter system.

Keywords: general relativity; stress-energy tensor; toroidal transport; helical transport; nonlinear stability; null energy condition; weak-field gravity

Classification of claims

Category	Meaning in this paper	Examples
Derived result	Obtained from the stated action and assumptions.	Field equations; stress-energy tensor; finite branch; stability; NEC theorem.
Numerical experiment	Computed in a reduced or weak-field model.	Dynamic collapse toy model; parameter scans; linearized cylindrical solve.
Hypothesis	Proposed behavior requiring nonlinear verification.	Persistence of the DEE; finite curvature; global non-collapse.

I. Introduction

Gravitational collapse is ordinarily analyzed by specifying a matter model, solving the Einstein equations, and determining whether trapped surfaces, horizons, or singular behavior form. A recurring difficulty in speculative non-collapse constructions is that positive separation is imposed directly through the geometry. Such a construction can demonstrate non-intersection by design but cannot establish that separation emerges from dynamics.

The objective of this work is narrower and testable. We ask whether a radial collapse degree of freedom can be coupled to directed toroidal and helical transport so that a finite equilibrium branch appears as a solution of the field equations of an explicit matter model. We refer to the transition into such a state as a Dynamic Equilibrium Event. The term 'event' denotes a dynamical transition, not a new causal boundary.

Two transport regimes are distinguished. The first is a nonlinear radial response, represented by a scalar field whose effective force changes with its local state. The second is linear non-waveform transport: directed flow represented by constant or slowly varying field gradients rather than by a sinusoidal carrier. 'Linear' therefore describes the transport profile, not the complete Einstein-matter equations, which remain nonlinear.

The manuscript proceeds from a covariant action, derives the matter equations and stress tensor, solves a minimal finite-separation branch, and proves a null-energy condition for the matter sector. An explicit localized configuration is then used as a source for the linearized cylindrical Einstein equations. Earlier exploratory metric ansätze are retained only as diagnostic calculations, because they reveal how unsupported off-diagonal geometry may demand inadmissible stress-energy.

The principal contribution is therefore not a claim of completed nonlinear non-collapse. It is an internally consistent weak-field framework that identifies the mathematical conditions a full nonlinear DEE solution must satisfy.

A. Main contributions

- A covariant three-field action separating radial nonlinear response from toroidal and helical transport.
- Closed-form existence and local-stability conditions for a finite-separation DEE branch.
- A positive-definite kinetic-matrix criterion that gives an analytical NEC proof for every null direction.
- An explicit non-waveform transport configuration with detailed angular energy flux and stress.
- A localized, linearized cylindrical Einstein solve satisfying the weak-field requirement after positive rescaling.
- A precise nonlinear verification program based on constraint-satisfying initial data, curvature invariants, and full null-cone testing.

B. Scope and limitations

This work does not present a global solution of the full nonlinear Einstein-matter equations. It does not prove avoidance of trapped surfaces or singularities, and it does not establish experimental realizability. The numerical values reported below are normalized exploratory values and require independent reproduction with a documented numerical code before journal submission. The weak-field solution should be understood as a consistency checkpoint and initial-data seed.

II. Conventions and geometric framework

We use metric signature (-,+,+,+) and units c=1. Newton's constant G is retained where useful. Greek indices run over spacetime coordinates, and hatted Latin indices denote components in a local orthonormal frame. Repeated indices are summed.

$$x^{\mu} = (t, r, \varphi, z). \quad (1)$$

The reduced cylindrical coordinates are used to develop and test the local mechanism. A global toroidal embedding would require periodic identification and additional regularity analysis; that extension is not assumed in the derivations below.

The radial scalar ρ(x) represents the finite-separation or collapse coordinate. The phase fields Θ(x) and Ψ(x) represent toroidal and helical transport channels, respectively. The term phase does not require periodic wave propagation; only the gradients enter the action.

A. Dynamic Equilibrium Event

A DEE is defined locally by a finite radial value at which the radial field is stationary and stable under small perturbations.

$$\rho = \rho_* > 0, \quad \nabla_t \rho|_* = 0, \quad \nabla_t\nabla_t \rho|_* = 0. \quad (2)$$

For a homogeneous local reduction, stability requires a positive effective radial mass squared. In the full field problem this condition is necessary but not sufficient for global stability.

III. Covariant action and field equations

We consider Einstein gravity coupled to one radial scalar and two kinetically mixed transport scalars.

$$S = \int d^4x \sqrt{(-g)} [R/(16\pi G) - 1/2 (\nabla\rho)^2 - V(\rho) - 1/2 K_\Theta(\rho)(\nabla\Theta)^2 - 1/2 K_\Psi(\rho)(\nabla\Psi)^2 - \Lambda(\rho) \nabla_\mu\Theta \nabla^\mu\Psi]. \quad (3)$$

The functions K_Θ, K_Ψ, and Λ determine the transport response and kinetic mixing. This action is an ansatz: its consequences are derived exactly, but its microscopic origin is not specified.

A. Radial equation

$$\Box\rho - V'(\rho) - 1/2 K_\Theta'(\rho)(\nabla\Theta)^2 - 1/2 K_\Psi'(\rho)(\nabla\Psi)^2 - \Lambda'(\rho)\nabla_\mu\Theta\nabla^\mu\Psi = 0. \quad (4)$$

The transport terms therefore modify the radial force through derivatives of the coupling functions. No centrifugal barrier is inserted separately.

B. Transport equations and currents

$$\nabla_\mu [K_\Theta(\rho)\nabla^\mu\Theta + \Lambda(\rho)\nabla^\mu\Psi] = 0, \quad (5)$$

$$\nabla_\mu [K_\Psi(\rho)\nabla^\mu\Psi + \Lambda(\rho)\nabla^\mu\Theta] = 0. \quad (6)$$

$$J_\Theta^\mu = K_\Theta\nabla^\mu\Theta + \Lambda\nabla^\mu\Psi, \quad J_\Psi^\mu = K_\Psi\nabla^\mu\Psi + \Lambda\nabla^\mu\Theta. \quad (7)$$

Equations (5)-(7) define conserved directed transport currents. They permit monotonic transport profiles and do not require sinusoidal waveform solutions.

IV. Stress-energy tensor and angular stresses

Variation of the matter action with respect to the inverse metric gives

$$T_{\mu\nu} = \nabla_\mu\rho\nabla_\nu\rho + K_\Theta\nabla_\mu\Theta\nabla_\nu\Theta + K_\Psi\nabla_\mu\Psi\nabla_\nu\Psi + \Lambda(\nabla_\mu\Theta\nabla_\nu\Psi + \nabla_\mu\Psi\nabla_\nu\Theta) + g_{\mu\nu} L_m, \quad (8)$$

$$L_m = -1/2(\nabla\rho)^2 - V - 1/2K_\Theta(\nabla\Theta)^2 - 1/2K_\Psi(\nabla\Psi)^2 - \Lambda\nabla_\alpha\Theta\nabla^\alpha\Psi. \quad (9)$$

The sign in the final term of Eq. (8) follows from the definition $T_{\mu\nu} = -2 \delta(\sqrt{-g} L_m)/(\sqrt{-g} \delta g^{\mu\nu})$.

A. Local angular decomposition

$$T_{\hat{a}\hat{b}} = [[\varepsilon, q_r, q_\varphi, q_z], [q_r, p_r, \pi_{r\varphi}, \pi_{rz}], [q_\varphi, \pi_{r\varphi}, p_\varphi, \pi_{\varphi z}], [q_z, \pi_{rz}, \pi_{\varphi z}, p_z]]. \quad (10)$$

The quantities most relevant to the counter-rotating null channel are the angular energy flux q_φ, azimuthal pressure p_φ, and radial-angular shear $\pi_{r\varphi}$.

$$q_\varphi = T_{\hat{t}\hat{\varphi}}, \quad p_\varphi = T_{\hat{\varphi}\hat{\varphi}}, \quad \pi_{r\varphi} = T_{\hat{r}\hat{\varphi}}, \quad \pi_{\varphi z} = T_{\hat{\varphi}\hat{z}}. \quad (11)$$

B. Directional null-energy condition

For the co-rotating and counter-rotating local null vectors $k_{\pm}^{\hat{a}} = (1,0,\pm1,0)$, the null contraction is

$$N_{\pm} = T_{\hat{a}\hat{b}}\, k_{\pm}^{\hat{a}}\, k_{\pm}^{\hat{b}} = \varepsilon + p_{\varphi} \pm 2q_{\varphi}. \quad (12)$$

$$\varepsilon + p_{\varphi} \geq 2|q_{\varphi}| \quad (13)$$

Equation (13) is necessary and sufficient for the NEC in both purely azimuthal null directions. It is not by itself sufficient for every null direction.

V. Minimal solvable DEE model

To obtain an explicit finite branch, we choose the regular polynomial functions

$$V(\rho) = \lambda_V/4\, (\rho^2 - v^2)^2, \quad (14)$$

$$K_{\Theta}(\rho) = 1 + a\rho^2, \quad K_{\Psi}(\rho) = 1 + b\rho^2, \quad \Lambda(\rho) = c\rho^2. \quad (15)$$

Here $\lambda_V>0$, $v>0$, and a,b,c are dimensionful or dimensionless according to the normalization of ρ. In a complete dimensional treatment, a reference mass scale must be restored.

$$X_{\Theta}=(\nabla\Theta)^2, \quad X_{\Psi}=(\nabla\Psi)^2, \quad X_{\Theta\Psi}=\nabla_{\mu}\Theta\nabla^{\mu}\Psi. \quad (16)$$

A. Finite-separation branch

$$\rho_* \left[\lambda_V(\rho_*^2-v^2) + aX_{\Theta} + bX_{\Psi} + 2cX_{\Theta\Psi} \right] = 0. \quad (17)$$

$$\rho_*^2 = v^2 - [aX_{\Theta}+bX_{\Psi}+2cX_{\Theta\Psi}]/\lambda_V. \quad (18)$$

The collapse branch is $\rho_*=0$. A real positive finite branch exists when the right-hand side of Eq. (18) is strictly positive.

$$aX_{\Theta}+bX_{\Psi}+2cX_{\Theta\Psi} < \lambda_V v^2. \quad (19)$$

B. Radial stability

$$m_{eff}^2 = V''(\rho_*) + 1/2K_{\Theta}''X_{\Theta} + 1/2K_{\Psi}''X_{\Psi} + \Lambda''X_{\Theta\Psi}. \quad (20)$$

$$m_{eff}^2 = 2\lambda_V \rho_*^2 > 0 \quad \text{for } \rho_*>0. \quad (21)$$

Thus every positive equilibrium in this minimal homogeneous reduction is locally stable in the radial scalar direction. Coupled metric and inhomogeneous perturbations remain to be tested.

C. Ghost-free kinetic sector

$$K(\rho) = [[K_{\Theta}, \Lambda], [\Lambda, K_{\Psi}]]. \quad (22)$$

$$K_{\Theta}>0, \quad K_{\Psi}>0, \quad \det K = K_{\Theta}K_{\Psi}-\Lambda^2 > 0. \quad (23)$$

$$\det K = 1+(a+b)\rho^2+(ab-c^2)\rho^4. \quad (24)$$

A simple sufficient condition for global positivity in this polynomial family is $a\geq0$, $b\geq0$, and $c^2<ab$.

VI. Explicit transport configurations and an NEC theorem

We choose non-oscillatory linear-gradient configurations in a local cylindrical region:

$$\Theta = \omega_{\Theta}\, t + m_{\Theta}\, \varphi + k_{\Theta}\, z, \quad (25)$$

$$\Psi = \omega_{\Psi}\, t + m_{\Psi}\, \varphi + k_{\Psi}\, z. \quad (26)$$

Single-valued scalar phases on a globally periodic angular coordinate would normally require integer winding numbers or an angular-valued field interpretation. Here m_{Θ} and m_{Ψ} are local winding parameters.

$$\partial_{\hat{a}}\Theta = (\omega_{\Theta},0,m_{\Theta}/r,k_{\Theta}), \quad \partial_{\hat{a}}\Psi = (\omega_{\Psi},0,m_{\Psi}/r,k_{\Psi}). \quad (27)$$

Let k^â=(1,n_r,n_φ,n_z), with n_r^2+n_φ^2+n_z^2=1. Define

$$U_\Theta = k^\hat{a}\partial_\hat{a}\Theta, \quad U_\Psi = k^\hat{a}\partial_\hat{a}\Psi. \quad (28)$$

$$T_{\hat{a}\hat{b}}\, k^\hat{a}\, k^\hat{b} = K_\Theta\, U_\Theta^2 + K_\Psi\, U_\Psi^2 + 2\Lambda U_\Theta U_\Psi. \quad (29)$$

$$T_{kk} = K_\Theta(U_\Theta + \Lambda U_\Psi/K_\Theta)^2 + (K_\Psi - \Lambda^2/K_\Theta)U_\Psi^2. \quad (30)$$

The metric-proportional Lagrangian term vanishes in the null contraction. Equations (23) and (30) therefore give the following result.

Proposition 1. For the matter action in Eq. (3), if the transport kinetic matrix is positive definite, then $T_{\mu\nu} k^\mu k^\nu \geq 0$ for every null vector k^μ. The radial scalar contributes an additional nonnegative square.

A. Illustrative normalized point

For λ_V=4, v=1, a=0.30, b=0.20, c=0.10, X_Θ=0.60, X_Ψ=0.40, and X_ΘΨ=-0.20, Eq. (18) gives ρ_*^2=0.945 and ρ_*≈0.972. The corresponding kinetic coefficients are K_Θ≈1.2835, K_Ψ≈1.189, and Λ≈0.0945, with det K≈1.52>0. This point is illustrative and not a fitted physical parameter set.

VII. Exploratory reduced-model studies

The calculations in this section are numerical experiments. They motivated the action-derived construction but are not used as proofs.

A. Dynamic radial-collapse toy model

$$\ddot{\rho} = -\gamma\dot{\rho} - \mu/(\rho+\varepsilon_0)^2 + L^2/(\rho+\varepsilon_0)^3. \quad (31)$$

In normalized integrations, the L=0 trajectory reached a small collapse threshold, while positive L generated a finite-radius barrier. This established only that rotational support can oppose collapse in a mechanics toy model; it did not derive the support from general relativity.

B. Prescribed-metric diagnostics

Several reduced off-diagonal cylindrical metrics were tested by computing the inferred Einstein tensor and directly sampling $T_{\mu\nu}k^\mu k^\nu$. Those trials frequently violated the NEC, particularly for counter-rotating azimuthal null directions near the inner radius during the DEE switch-on. The diagnostic conclusion was that an off-diagonal metric should not be prescribed independently of a compatible matter source.

A parameter search localized the most negative sampled null direction near the transition and primarily opposite the angular transport. This observation motivated the explicit matter-sector derivation and Eq. (13).

VIII. Linearized Einstein solution

We now solve for the weak gravitational field produced by the explicit matter model. Write

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (32)$$

In harmonic gauge, the trace-reversed perturbation satisfies

$$\Box\, \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}. \quad (33)$$

For a stationary, translationally invariant cylindrical approximation with radial dependence only,

$$(1/r)\, d/dr\, [\, r\, d\, \bar{h}_{\mu\nu}/dr\,] = -16\pi G\, T_{\mu\nu}(r). \quad (34)$$

$$d\, \bar{h}_{\mu\nu}/dr = -(16\pi G/r) \int_0^r s\, T_{\mu\nu}(s)\, ds. \quad (35)$$

A Gaussian radial envelope is applied to the explicit field-gradient source to obtain a localized finite computational domain. This envelope is a numerical modeling choice and is not derived from Eq. (3); a self-consistent localized nonlinear field solution remains future work.

A. Source normalization and weak-field checkpoint

The initial normalized source generated perturbations larger than unity, invalidating the linear approximation. Because Eqs. (33)-(35) are linear in the source, we introduced a positive overall amplitude ε_s.

$$T_{\mu\nu} \to \varepsilon_s T_{\mu\nu}, \quad h_{\mu\nu} \to \varepsilon_s h_{\mu\nu}. \quad (36)$$

$$\varepsilon_s = 10^{-6}. \quad (37)$$

Diagnostic	Scaled value	Status
Largest $\lvert h_{\mu\nu}\rvert$	1.16×10^{-3}	Weak-field pass
Minimum co-rotating N_+	3.71×10^{-14}	Positive
Minimum counter-rotating N_-	1.69×10^{-14}	Positive
Minimum sampled energy density	2.62×10^{-14}	Positive

These values constitute the linearized Einstein-DEE consistency checkpoint. The reported digits reflect the exploratory computation and should not be interpreted as precision results.

Linearized Einstein-DEE consistency test: PASS. (38)

B. Meaning of the pass

The pass means that an explicit positive-definite matter source can generate a small cylindrical metric perturbation while maintaining positive tested directional NEC contractions. It does not prove that the result persists under full nonlinear evolution.

IX. Nonlinear initial-value problem

The weak-field result supplies a seed for a nonlinear initial-data construction. On an initial spacelike slice,

$$g_{\mu\nu}(t_0,r) = \eta_{\mu\nu} + h_{\mu\nu}(r), \quad (39)$$

$$\partial_t g_{\mu\nu}(t_0,r) = \dot{A}_{DEE}(t_0)\, h_{\mu\nu}^{(DEE)}(r), \quad (40)$$

$$\rho(t_0,r)=\rho_*+\delta\rho(r), \quad \partial_t\rho(t_0,r)=\delta\dot{\rho}(r), \quad (41)$$

with corresponding data for Θ and Ψ. These data must be adjusted or solved so that the Hamiltonian and momentum constraints hold.

$$^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi G\, E, \quad (42)$$

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi G\, S^i. \quad (43)$$

A. Nonlinear pass criteria

$$\rho_{min}(t) = \min_r \rho(t,r) > 0, \quad (44)$$

$$N_{min}(t,r) = \min_{\{k \text{ null}\}} T_{\mu\nu} k^\mu k^\nu \geq 0, \quad (45)$$

$$R, \quad R_{\mu\nu}R^{\mu\nu}, \quad R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} \text{ remain finite}, \quad (46)$$

constraint residuals remain convergent under resolution refinement. (47)

Only after these conditions are demonstrated over a stated simulation interval can the finite-separation hypothesis be promoted to a nonlinear numerical result.

X. Discussion

The development above separates three questions that are often conflated. First, can a finite radial branch exist? Equation (18) answers yes within the model. Second, can that branch be locally stable? Equation (21) answers yes

in the homogeneous radial sector. Third, can the supporting matter satisfy the NEC? Proposition 1 answers yes whenever the kinetic matrix is positive definite.

Those statements do not settle the gravitational question. The Einstein equations can generate additional instabilities, constraint violations, trapped surfaces, or curvature growth. The weak-field calculation shows that there is no immediate inconsistency at sufficiently small source amplitude, but the nonlinear regime is where the central hypothesis must be tested.

The distinction between nonlinear radial response and linear non-waveform transport is useful because it assigns different roles to the degrees of freedom. The radial scalar determines the local separation branch, while the transport gradients carry angular and helical momentum. Their coupling changes the radial effective force without requiring an independently postulated centrifugal term.

A. Relation to angular stress

In the local orthonormal frame, the counter-rotating null direction is controlled by $\varepsilon+p_\varphi-2q_\varphi$. A prescribed metric shift can make q_φ too large compared with the supporting energy and pressure. In the action-derived model, the full null contraction is instead the quadratic form in Eq. (29), and positive definiteness of the kinetic matrix protects every null direction.

B. Falsifiability

The framework can fail in several concrete ways: the initial constraints may have no regular solution in the desired parameter region; the finite branch may destabilize under coupled metric perturbations; the solution may develop trapped surfaces or divergent curvature; or the NEC may fail after nonlinear backreaction. Any of these outcomes would falsify the strong non-collapse hypothesis for the tested model.

XI. Status of results

Result	Classification	Status
Field equations from Eq. (3)	Derived	Complete
Stress-energy tensor	Derived	Complete
Finite branch Eq. (18)	Derived	Complete within minimal model
Radial stability Eq. (21)	Derived	Local homogeneous result
NEC proposition	Derived	Complete for the stated matter action
Dynamic collapse and metric scans	Numerical experiment	Exploratory
Linearized cylindrical solution	Numerical experiment	Weak-field consistency pass
Full nonlinear finite-separation evolution	Hypothesis	Not yet tested
Global absence of singularities	Hypothesis	Not established

XII. Conclusion

We have constructed a covariant coupled-field model in which a radial nonlinear degree of freedom interacts with toroidal and helical transport gradients. A minimal polynomial realization admits a closed-form finite branch, and every positive point on that branch is locally stable in the radial scalar sector. The matter stress tensor satisfies the NEC for all null directions whenever its kinetic matrix is positive definite. An explicit localized source produces a consistent cylindrical linearized Einstein response after a positive amplitude rescaling, with metric perturbations of order 10^{-3} and positive tested co- and counter-rotating directional NEC contractions.

The present work therefore establishes a mathematically coherent weak-field DEE framework. It does not establish global non-collapse. The next decisive calculation is a constraint-preserving nonlinear Einstein-matter

evolution that monitors the minimum separation, full null cone, curvature invariants, trapped-surface diagnostics, and convergence of the constraints.

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The author used computational and language-model assistance during conceptual development, algebra checking, numerical prototyping, and manuscript organization. All scientific claims, interpretations, and responsibility for the submitted work remain with the author. The author should replace this statement as required by the target journal's current disclosure policy.

Data and code availability

The exploratory numerical calculations described in this draft should be consolidated into a version-controlled repository before external circulation. A submission version should provide the complete source code, parameter files, software versions, convergence tests, and machine-readable output used to generate every numerical result.

Conflict of interest

The author declares no conflict of interest.

Appendix A: Variation of the action

This appendix records the essential variations. For a scalar f, δ[(∇f)^2]=δg^{μν}∇_μf∇_νf. The kinetic-mixing term varies symmetrically in Θ and Ψ. Integrating by parts and discarding boundary terms gives Eqs. (4)-(6).

$$\delta S/\delta\rho = 0 \Rightarrow \text{Eq. (4)}, \quad \delta S/\delta\Theta = 0 \Rightarrow \text{Eq. (5)}, \quad \delta S/\delta\Psi = 0 \Rightarrow \text{Eq. (6)}. \quad \text{(A1)}$$

Metric variation gives Eq. (8). Before submission, the author should include the full line-by-line derivation and verify all sign conventions against the chosen definition of the Einstein tensor.

Appendix B: Proof of the NEC proposition

For a null vector k^μ, the metric-proportional term in Eq. (8) vanishes. Define u=(k·∇Θ, k·∇Ψ)^T. The transport contribution is u^T K u. The radial scalar contributes (k·∇ρ)^2. If K is positive semidefinite, both terms are nonnegative; if K is positive definite, the transport term vanishes only when both projected gradients vanish.

$$T_{\mu\nu}k^\mu k^\nu = (k\cdot\nabla\rho)^2 + u^T K\, u \geq 0. \quad \text{(B1)}$$

Appendix C: Linearized cylindrical solver

For each component, Eq. (34) was integrated through the enclosed radial source. With a finite outer boundary R_max and boundary condition $\bar{h}_{\mu\nu}(R_{max})=0$, the formal solution is

$$\bar{h}_{\mu\nu}(r) = 16\pi G \int_r^{R_{max}} [1/s] \left[\int_0^s u\, T_{\mu\nu}(u)\, du\right] ds. \quad \text{(C1)}$$

The numerical implementation used finite radial sampling and cumulative quadrature. The present draft reports only summary diagnostics. A referee version requires grid-refinement studies and independent residual evaluation of Eq. (34).

Appendix D: Symbol glossary

Symbol	Meaning
ρ	Radial separation/collapse scalar
Θ	Toroidal transport phase field
Ψ	Helical transport phase field
V(ρ)	Radial potential
K_Θ, K_Ψ	Field-dependent transport kinetic coefficients
Λ	Toroidal-helical kinetic mixing
ρ_*	Finite DEE equilibrium value
m_eff^2	Local radial perturbation mass squared
q_φ	Azimuthal energy flux
p_φ	Azimuthal pressure
π_rφ	Radial-angular shear stress
N_±	Co- and counter-rotating azimuthal null contractions
h_μν	Linear metric perturbation
ε_s	Positive source-amplitude scaling parameter

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Internal referee checklist for Version 1.1

- ☐ Verify every sign in Eqs. (3)-(9) using a single stated convention.
- ☐ Restore physical dimensions and introduce a reference mass/length scale.
- ☐ Resolve global angular-field single-valuedness and winding-number conditions.
- ☐ Provide complete numerical source code and reproducible environment.
- ☐ Perform convergence and residual tests for the linearized solver.
- ☐ Construct constraint-satisfying nonlinear initial data.
- ☐ Add trapped-surface diagnostics to the nonlinear evolution plan.
- ☐ Replace placeholder ORCID, email, DOI list, and repository references.
- ☐ Obtain an independent technical read by a general-relativity specialist.
- ☐ Update journal-format and AI-assistance disclosures immediately before submission.